

Linear asymptotics for the L_2 regularity of tile B-splines

Candidate partial solution packet for arXiv:2212.12945

13 August 2026

Status. Candidate partial result, likely valid. The result proves a general linear asymptotic law, monotonicity, concavity, and slope bounds for the L_2 regularity. It does not compute the limiting slope for the Dragon or Bear, and it does not address C -regularity.

1 Source question

The source is T. I. Zaitseva, *Multivariate tile B-splines*, arXiv:2212.12945, later published in *Izvestiya: Mathematics* 87(2) (2023), 284–325 [1]. If $G \subset \mathbb{R}^d$ is a tile, the tile B-spline of order ℓ is

$$B_\ell^G = \underbrace{\chi_G * \cdots * \chi_G}_{\ell+1 \text{ factors}}.$$

After computing small-order regularities by growing transition matrices, the paper asks in Remark 6 (PDF page 33):

“The question of the asymptotics of the regularity with the increase of order of B-splines remains open.”

Remark 6 *The values of regularity for large orders of tile B-splines are not given since the computation of joint spectral radius is hard for large sizes of matrices. The transition matrices grow fast when we increase the order of convolution. The question of the asymptotics of the regularity with the increase of order of B-splines remains open.*

Figure 1: The complete source question, rendered from page 33 of arXiv:2212.12945.

The source considers both C -regularity and L_2 -Hölder regularity. The result below addresses the latter. It applies more generally to convolution powers of any normalized bounded measurable set.

2 Proof intuition

The growing refinement matrices disappear after taking Fourier transforms: if $F = \widehat{\chi_G}$, then $\widehat{B_\ell^G} = F^{\ell+1}$. Regularity is therefore the boundary of the set of pairs (p, s) for which

$$\int_{|\xi| \geq 1} |\xi|^{2s} |F(\xi)|^{2p} d\xi < \infty.$$

Hölder’s inequality makes this integrability set convex, so its upper boundary as a function of p is concave. Its value at $p = 0$ is exactly $-d/2$. A nonnegative concave function vanishing at the origin has

decreasing secant slopes; this yields a limiting regularity per convolution and exact finite-order upper bounds on it.

The tile sum rules bound the limiting slope above by 1. In the other direction, a p th difference of a p fold convolution is the convolution of p first differences. Young's inequality converts the translation modulus of the tile itself into the lower bound $L_G \geq 2\alpha_{0,2}(G)$. What remains open is the computation of L_G for particular fractal tiles and the analogous C -regularity problem.

3 Definitions and a threshold lemma

Let $G \subset \mathbb{R}^d$ be bounded and measurable with $|G| = 1$, put $f = \chi_G$, and use the Fourier convention $\widehat{f}(\xi) = \int f(x)e^{-2\pi i x \cdot \xi} dx$. Then $F = \widehat{f}$ satisfies $|F| \leq 1$. For real $p \geq 0$, define the extended regularity threshold

$$A_G(p) = \sup \left\{ s \in \mathbb{R} : \int_{|\xi| \geq 1} |\xi|^{2s} |F(\xi)|^{2p} d\xi < \infty \right\}, \quad (1)$$

where $|F|^0 = 1$.

For a compactly supported $g \in L_2(\mathbb{R}^d)$, let $\alpha_2(g)$ denote the supremal L_2 -Hölder/Besov regularity used by the source. Only the supremal threshold matters here, so endpoint conventions at integral exponents do not affect the result.

Lemma 1 (Sobolev and L_2 -Hölder thresholds). *For compactly supported $g \in L_2(\mathbb{R}^d)$,*

$$\alpha_2(g) = \sup \{s : g \in H^s(\mathbb{R}^d)\}.$$

Consequently,

$$\alpha_{\ell,2}(G) := \alpha_2(B_\ell^G) = A_G(\ell + 1). \quad (2)$$

Proof. Let

$$E_j = \int_{2^j \leq |\xi| < 2^{j+1}} |\widehat{g}(\xi)|^2 d\xi, \quad j \geq 0.$$

Up to the harmless low-frequency term, membership in H^s is equivalent to $\sum_j 2^{2sj} E_j < \infty$, while the $B_{2,\infty}^t$ condition is $\sup_j 2^{2tj} E_j < \infty$. The former implies the latter at the same exponent. Conversely, the latter at exponent t implies the former at every $s < t$ by a geometric series. Their supremal thresholds are equal. Since $\widehat{B_\ell^G} = F^{\ell+1}$, the Sobolev integral gives (2). $\square$

4 The linear-asymptotic theorem

Theorem 2 (Concavity and asymptotic slope). *Let $G \subset \mathbb{R}^d$ be bounded, measurable, and $|G| = 1$. Assume $\alpha_{0,2}(G) < \infty$, as holds for every nontrivial tile indicator. Then:*

- (i) A_G is nondecreasing and concave on $[0, \infty)$, and $A_G(0) = -d/2$.
- (ii) The shifted normalized regularities

$$q_\ell(G) = \frac{\alpha_{\ell,2}(G) + d/2}{\ell + 1} \quad (3)$$

are nonincreasing. Hence

$$L_G := \lim_{\ell \rightarrow \infty} q_\ell(G) = \inf_{\ell \geq 0} q_\ell(G) \quad (4)$$

exists and is finite and nonnegative.

(iii) *The regularity has the first-order asymptotic*

$$\alpha_{\ell,2}(G) = L_G(\ell + 1) - \frac{d}{2} + o(\ell). \quad (5)$$

Here the displayed $-d/2$ is the natural concavity normalization, not a claim about a second-order constant.

(iv) *The sequence $\alpha_{\ell,2}(G)$ is nondecreasing and discretely concave. Its marginal gains decrease and satisfy*

$$\alpha_{\ell+1,2}(G) - \alpha_{\ell,2}(G) \downarrow L_G. \quad (6)$$

If G is a lattice tile and its tile B-splines are stable with the exact sum-rule orders described in [1, 2], then

$$0 \leq L_G \leq 1. \quad (7)$$

Proof. Suppose the integral in (1) is finite at (p_0, s_0) and (p_1, s_1) . For $0 < \theta < 1$, set

$$p_\theta = \theta p_0 + (1 - \theta)p_1, \quad s_\theta = \theta s_0 + (1 - \theta)s_1.$$

The integrand at (p_θ, s_θ) is the product of the two endpoint integrands raised to the powers θ and $1 - \theta$. Hölder's inequality proves its integrability. Letting $s_i \uparrow A_G(p_i)$ shows

$$A_G(p_\theta) \geq \theta A_G(p_0) + (1 - \theta)A_G(p_1),$$

so A_G is concave. Since $|F| \leq 1$, increasing p can only decrease the integrand, hence A_G is nondecreasing. At $p = 0$, radial integration gives

$$\int_{|\xi| \geq 1} |\xi|^{2s} d\xi < \infty \iff s < -d/2,$$

and therefore $A_G(0) = -d/2$.

Put $C(p) = A_G(p) + d/2$. Then C is nonnegative and concave, with $C(0) = 0$. For $0 < p < r$, concavity gives

$$C(p) \geq \frac{p}{r} C(r),$$

so $C(p)/p$ is nonincreasing. Restricting to $p = \ell + 1$ and using Lemma 1 proves (3)–(5). The assumed finiteness at $p = 1$ and the secant inequality make the limit finite. Monotonicity and concavity of A_G restricted to the integers give the corresponding properties of $\alpha_{\ell,2}$. The decreasing discrete increments have a limit, and the Cesàro mean identity forces that limit to equal L_G , proving (6).

Finally, the source proves stability of tile B-splines and identifies their sum-rule order. The standard stability/sum-rule obstruction recalled and used in [2] gives $\alpha_{\ell,2}(G) \leq \ell + 1$. Divide by $\ell + 1$ and take the limit to obtain $L_G \leq 1$; nonnegativity was already proved. $\square$

Corollary 3 (A geometric lower bound). *For every normalized tile G ,*

$$2\alpha_{0,2}(G) \leq L_G \leq 1. \quad (8)$$

In particular, the square has $L_G = 1$.

Proof. Write $f = \chi_G$, $D(h) = |G \triangle (G + h)|$, and $\Delta_h = \tau_h - I$. Then

$$\|\Delta_h f\|_1 = D(h), \quad \|\Delta_h f\|_2 = D(h)^{1/2}. \quad (9)$$

For an integer $p \geq 1$, the convolution and translation algebras give

$$\Delta_h^p(f^{*p}) = (\Delta_h f)^{*p}. \quad (10)$$

Young's inequality and (9) imply

$$\|\Delta_h^p(f^{*p})\|_2 \leq \|\Delta_h f\|_2 \|\Delta_h f\|_1^{p-1} = D(h)^{p-1/2}. \quad (11)$$

Fix $0 \leq \eta < \alpha_{0,2}(G)$. By the definition of the regularity of χ_G , $D(h)^{1/2} \leq C_\eta |h|^\eta$ for small h . Thus

$$\|\Delta_h^p(f^{*p})\|_2 \leq C_{\eta,p} |h|^{2\eta(p-1/2)}.$$

A nonconstant compactly supported indicator has $\alpha_{0,2} \leq 1/2$, so $2\eta(p-1/2) < p$. The p th-difference characterization of $B_{2,\infty}^s$ therefore yields

$$\alpha_{p-1,2}(G) \geq 2\eta(p-1/2).$$

Divide by p , let $p \rightarrow \infty$, and then let $\eta \uparrow \alpha_{0,2}(G)$ to get $L_G \geq 2\alpha_{0,2}(G)$. The upper bound is Theorem 2. For the square, $\alpha_{0,2} = 1/2$, so equality follows. $\square$

5 Consequences for the source examples

In dimension two, every computed order gives the rigorous implication

$$L_G \leq \frac{\alpha_{\ell,2}(G) + 1}{\ell + 1}, \quad (12)$$

and these upper bounds must decrease with ℓ . The tables in [2] exhibit exactly this pattern. Using their displayed decimal values only as numerical evidence, not as interval-certified proof, the order-8 Bear value 8.9199 gives $L_{\text{Bear}} \lesssim 1.10221$, while the order-5 Dragon value 3.6688 gives $L_{\text{Dragon}} \lesssim 0.77813$. Corollary 3 and the source's order-zero decimals give the indicative lower bounds

$$L_{\text{Bear}} \gtrsim 0.7892, \quad L_{\text{Dragon}} \gtrsim 0.4764.$$

The exact theorem is the monotonicity and limiting law; the decimal endpoints inherit the precision limitations of the published tables.

6 Verification, limitations, and novelty status

Internal checks. The Fourier-power identity and the dyadic threshold equivalence were checked independently, as were all exponents in Hölder's and Young's inequalities. The theorem reproduces the square sequence $\alpha_{\ell,2} = \ell + 1/2$: its normalized quantities are $1 + 1/(2(\ell + 1))$, decreasing to 1. The published Dragon and Bear tables also have decreasing marginal gains and decreasing normalized shifted values, as the theorem requires.

Bounded literature audit. The search covered the source and its citations, exact question phrases, arXiv/title/author searches, the run indexes, arXiv:2312.11182 and its 2025 journal version [2], and Zaitseva's 2025 *Supersmooth tile B-splines* [3]. Those papers give new finite-order formulas and tables and a supersmoothness classification conjecture. No theorem asserting concavity, existence of the large-order slope, or the displayed geometric lower bound was found through 13 August 2026. This bounded search does not certify novelty.

Limitations. The result does not compute L_G for Dragon, Bear, or a general tile. It does not show that the remainder in (5) is bounded, and the $-d/2$ normalization is not a second-order asymptotic. It does not address the joint-spectral-radius asymptotics governing C -regularity. Hence this is a substantial partial answer rather than a full resolution of every reading of Remark 6.

Human-review recommendation. High priority. The core proof is short and matrix-free. A reviewer should focus on Lemma 1, the endpoint-independent interpretation of the source regularity, the exact sum-rule upper bound, and the use of the p th-difference characterization in Corollary 3.

References

- [1] T. I. Zaitseva, *Multivariate tile B-splines*, Izv. Math. 87(2) (2023), 284–325; arXiv:2212.12945.
- [2] V. Yu. Protasov and T. Zaitseva, *Anisotropic refinable functions and the tile B-splines*, Appl. Comput. Harmon. Anal. 75 (2025), 101727; doi:10.1016/j.acha.2024.101727; arXiv:2312.11182.
- [3] T. I. Zaitseva, *Supersmooth tile B-splines*, Sb. Math. 216(3) (2025), 333–356; doi:10.4213/sm10212e.